
(a)
AdamW
Muon
step
250
500
1000
2000
3000
5000
10000
2.78
2.76
2.49
2.50
2.75
2.49
d2
d1
d0
1.94
1.77
1.76
1.71
1.61
1.73
1.77
1.95
1.71
(b)
AdamW vs Muon
AdamW, other data order
Muon, other data order
JS (bits)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
2000
4000
6000
8000
10000
step
(c)
JS (bits)
4
3
2.5
2
1.75
1.5
1.4
1.3
training loss
(d)
AdamW
Muon
validation loss
4.0
3.5
3.0
2.5
2.0
1.5
0
2000
4000
6000
8000
10000
step